

Hyson Detector

The AI copilot for bank fraud analysts: detect, explain, decide

Microsoft Fabric · Azure OpenAI · Azure AI Foundry · Power BI

6×
more frauds caught

−47 %
alerts for analysts

67.7 %
of fraud value stopped

Day 0
alert, vs 11 days to report

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All Hyson figures are measured on the organisers' synthetic Tunisian bank in Microsoft Fabric (validation June–July 2026 unless stated). External benchmarks carry their source. Scale-ups are labelled *scenario*: they are orders of magnitude to discuss with a partner bank, not measurements.

01 Executive summary

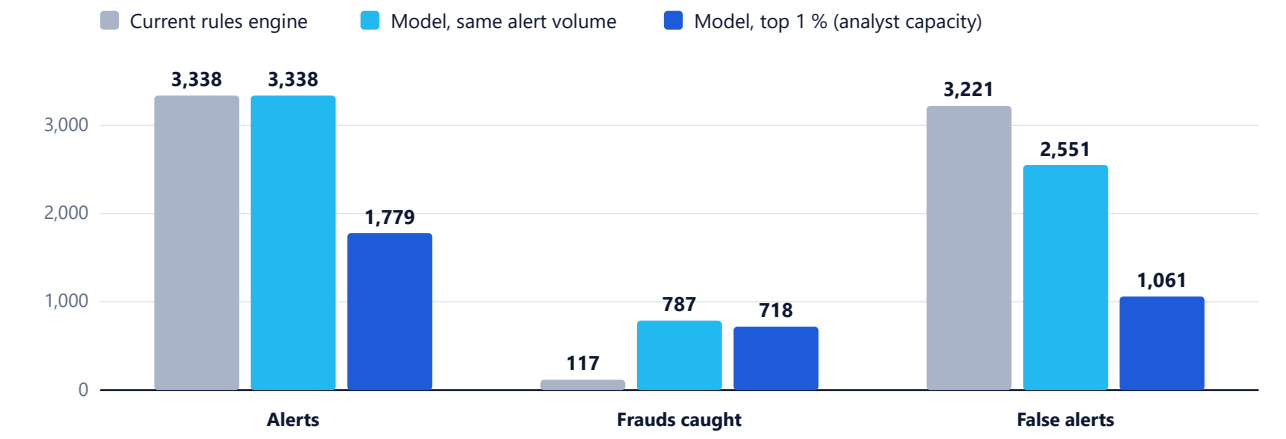
Hyson Detector is an AI copilot for the fraud analysts of a retail bank. It ranks every card and digital-banking operation by fraud risk, explains each alert with facts the analyst can check, answers questions in plain language, and prepares the decision (block, escalate or validate) that a named employee confirms. It runs on **Microsoft Fabric** (data, model, Power BI) and **Azure OpenAI / Azure AI Foundry** (copilot, assistant, investigation agents), and it was built during the DSI Forum hackathon on the organisers' synthetic Tunisian bank.

The problem, measured on the bank's own data. In June–July 2026 the current rules engine raised **3,338 alerts** and caught **117 of 1,013 frauds**: 3.5 % precision and 11.5 % recall. **96.5 % of the analysts' work went into false alerts**, and a fraud is reported a **median of 11 days** after it happens.

What Hyson changes. On the same two months, which the model never saw during training:

FRAUDS CAUGHT	PRECISION	RECALL	FALSE ALERTS AVOIDED
718	**40.4 %**	**70.9 %**	**2,160**
vs 117 for the current rules	vs 3.5 % · top 1 % of operations	vs 11.5 % · CI 68.3–73.0 %	1,061 left vs 3,221 today

FRAUD VALUE STOPPED	PR-AUC	STABLE OVER TIME	COPILOT ACCURACY
67.7 %	**0.646**	**69.4 %**	**18 / 18**
vs 22.1 % for the rules	vs 0.014 for the rules	recall ± 2.1 on 4 re-trainings	golden questions · 99.5 % grounded



Validation June–July 2026 (177,953 operations, 1,013 frauds). Top 1 %: 6.1× more frauds caught with 47 % fewer alerts.

Why it can be trusted.

- **It holds over time.** Re-trained four times on the past only, top-1 % recall stays at **69.4 % ± 2.1 points** (worst period 66.7 %). The 95 % confidence interval on recall is 68.3–73.0 %.
- **The score and its explanation come from one model.** LightGBM ranks; TreeSHAP on the same model gives three reasons for and two against each alert; a documented rule recommends; the AI only explains.
- **The AI never invents a number.** The copilot passes 18/18 golden questions and 99.5 % of the numbers it cites are found in the tool outputs. Reports reject any paragraph whose numbers are not in the evidence.

- **A human decides, and it is recorded.** Every decision, approval and report is written to a SHA-256 hash chain that anyone can verify.
- **It is fairer than today.** Between the most and least disturbed customer segments, false alerts differ by 1.4× with the model, against 2.8× with the current rules.

Beyond the score. Hyson checks whether an operation fits the customer's own life (a retired customer abroad is not automatically a fraud), keeps each case alive after the alert, writes evidence-based reports in French, English or Arabic, and plugs into any bank's data through a mapping a human approves.

The ask. A **3-month pilot with one bank**, retrained and validated on its real data, starting with one fraud team and its current alert volume.

02 The problem

2.1 Rules that drown analysts

Banks still rely on static rules that each look at one criterion: a large amount, a foreign country, a night payment. On the organisers' data we measured the current engine on the validation period (June–July 2026, 177,953 operations, 1,013 confirmed frauds):

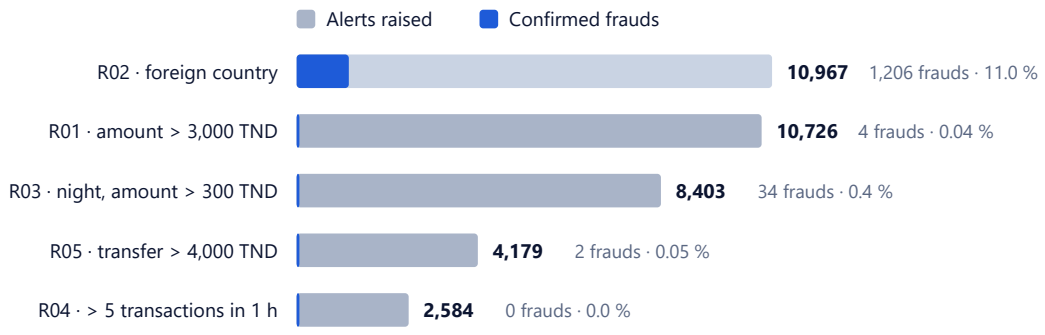
	Alerts	Frauds caught	False alerts	Precision	Recall
Current rules engine	3,338	117	3,221	3.5 %	11.5 %

Nine alerts out of ten waste an analyst's time, and nine frauds out of ten pass. The industry estimates that **90–95 %** of rule-based alerts are false positives (PwC / McKinsey, cited by Fluxforce, 2024); the engine we measured sits slightly above that range, at 96.5 %.

2.2 Which rules do the work

The legacy engine left 32,065 alerts with an analyst verdict. Auditing them rule by rule shows where the waste comes from:

Rule	Alerts	Confirmed frauds	Precision	Frauds it alone caught
R04 · more than 5 transactions in 1 h	2,584	0	0.0 %	0
R01 · amount above 3,000 TND	10,726	4	0.04 %	2
R05 · transfer above 4,000 TND	4,179	2	0.05 %	0
R03 · night, amount above 300 TND	8,403	34	0.4 %	32
R02 · foreign country	10,967	1,206	11.0 %	1,204



Audit of the current rules on 32,065 legacy alerts with analyst verdicts. R04, R01 and R05 together: 17,489 alerts, 6 frauds.

> **Quick win, before any AI:** switching off R04, R01 and R05 removes about **17,483 false alerts** for the loss of **2 frauds** that only they caught. R02 does almost all the legacy work and stays, alongside the model, as a complement for cloned cards used abroad.

2.3 Fraud is found too late

The bank records when each fraud is reported. The **median delay between the fraudulent operation and its report is 11 days**; only **32.8 %** of frauds are reported within 7 days, and **693,791 TND** of fraud was reported more than a week after the fact. By then the money has moved. Hyson scores operations as they arrive, so the alert exists at day 0.

2.4 Alerts without explanations

A rule says *what* fired, not *why it matters for this customer*. An analyst who sees "amount above 3,000 TND" must still open the customer's history, devices, beneficiaries and recent logins to decide. Without explanations, decisions are slow, inconsistent between analysts, and hard to justify to auditors and customers.

03 The solution: Hyson Detector

3.1 One principle

The model scores, the AI explains, the employee decides. Each layer has one job, and none does the job of another:

Layer	Does	Never does
Model (LightGBM in Fabric)	Ranks every operation by risk, with past-only signals	Block anything by itself
Rules and SHAP (code)	Turn the score into facts: amount vs habits, new device, new beneficiary, speed between two places	Guess
Decision rule (code)	Recommend BLOQUER / ESCALADER / VALIDER with a measured precision	Hide its confidence
Azure OpenAI (copilot, agents)	Explain, answer, summarise, argue, write	Compute a score or a figure, see raw rows, record a decision
Employee	Decide, override, comment	—

3.2 Detect → Explain → Decide → Learn

1. **Detect.** Every operation gets a fraud score; the riskiest 1 % (the analysts' capacity) becomes the alert queue, split into high and medium risk.
2. **Explain.** Each alert shows the observed value against the customer's usual value, the reassuring elements, and a recommendation backed by its measured precision.
3. **Decide.** The analyst asks the copilot, opens the investigation case, then blocks, escalates or validates. The decision, the system recommendation and any disagreement are logged.
4. **Learn.** Decisions return to OneLake as labels; the retraining pipeline replaces the model only if the new one wins on the same 4-period backtest.

3.3 A real case, step by step

Transaction **TX01555470**, a synthetic account takeover from the validation period:

- three failed logins in 24 hours from a new device with an Italian IP;
- a new beneficiary added;
- night transfers, the last one **4.3 times** the customer's usual amount.

The model places it in the **top 0.01 %**. The copilot explains it with these facts and recommends BLOQUER; the analyst confirms; the decision is hashed into the journal. Today this fraud would have been reported about 11 days later.

3.4 What is in the product

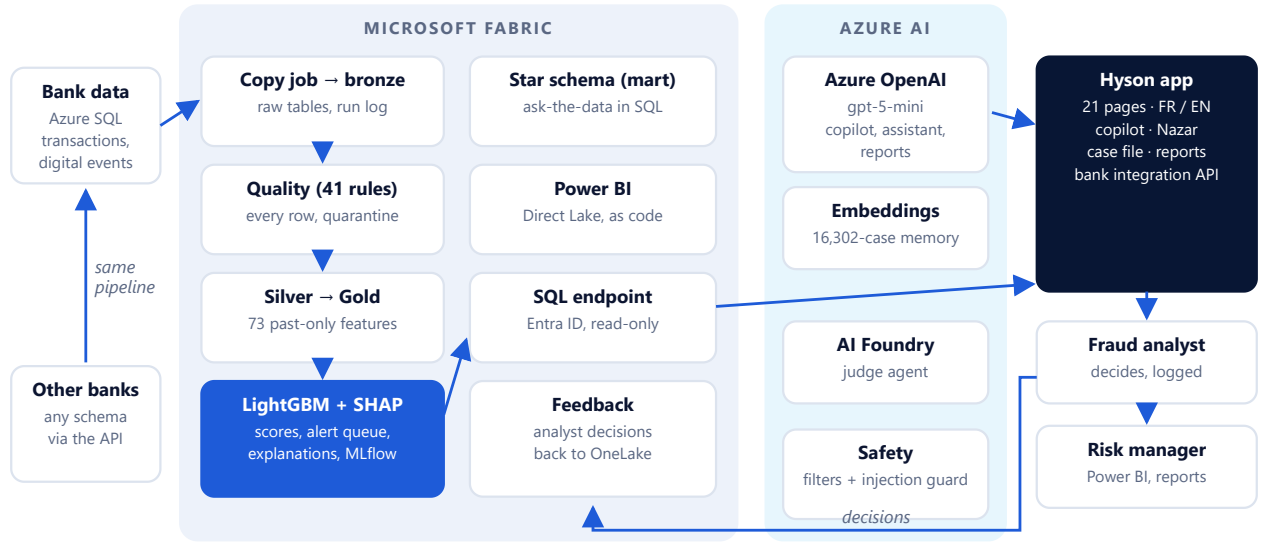
The app has **21 pages in 6 groups**, in French and English, with a floating agent (Nazar) on every page.

Group	Pages
Assistant	AI assistant: any question in natural language → guarded SQL → answer, chart, table

Group	Pages
Analyst	Alert queue · Transaction & copilot · Investigation case · Customer 360 · Personal context
AI investigation	Deep search · The Trial (prosecution, defence, judge) · Reconstitution (fraud-history roadmap) · Counter-inquiry (what-if) · Compromise network
Steering	Fraud intelligence · AI reports · Impact & process · Market & benchmarks · Model performance · Power BI · Audit log
Data	Bank integration (any schema) · New data source
Demo	Guided tour

04 Technology and architecture

4.1 End-to-end view



End-to-end architecture: data and model in Microsoft Fabric, language in Azure AI, decisions by people.

4.2 Microsoft Fabric data platform

- **Ingestion.** A Fabric copy job loads the bank's Azure SQL source into the lakehouse `lh_fraud`, logging rows, columns and a schema fingerprint on every run.
- **Medallion layout as code.** Schemas `bronze` (raw) → `quality` (41 rules) → `silver` (clean transactions) → `gold` (features, scores, alert queue, explanations) → `ml` (metrics, robustness) → `mart` (star schema) → `serving` (Power BI tables) → `meta` (catalogue, migration log). The map of every table lives in one file of the app (`layout.py`).
- **Row-by-row quality.** 41 documented rules run on **every row** of the 5 bronze tables (3.4 million records). Each failure is a row in `dq_issues`; each record gets OK / WARNING / REJECTED; rejected rows go to quarantine. Result on the delivered data: **0 rejected**, 11,756 warnings (for example 11,717 card-present operations without a place).
- **Star schema** for questions in plain language: `fact_transactions` (1.9 M), `fact_digital_events` (782 k) and seven dimensions (date with holidays and payday, hour, geography, customer without gender, merchant, channel, card), reconciled against bronze.
- **Power BI** semantic model in Direct Lake on the same lakehouse, built as code by a notebook and embedded in the app with Entra ID sign-in.
- **MLOps.** MLflow experiment and registered model `hyson-fraud-lightgbm` v1 (reproduces production scores exactly), data pipeline `pl_hyson_retrain`, library versions pinned.

4.3 Azure AI layer

Component	Model / service	Used for
Copilot, assistant, reports, Nazar chat	Azure OpenAI `gpt-5-mini` (reasoning effort low, capped output)	Explanations, natural-language SQL, report writing

Component	Model / service	Used for
Investigation agents	Azure OpenAI deployment for the prosecution and defence roles	Arguing the same evidence from both sides
Judge	Azure AI Foundry agent `judge-patern`	Grading each argument's evidence
Behavioural memory	`text-embedding-3-small` (256 dimensions)	Finding the 8 most similar past cases among 16,302

Azure content safety filters stay on; a prompt-injection guard cleans every tool result before it reaches a model; every call is logged with its tokens.

4.4 Application and delivery

- **Streamlit** app (FR / EN), live on the Fabric SQL endpoint with the analyst's Entra login, and a committed parquet snapshot for the hosted demo and the tests.
- **Universal Banking API** (FastAPI) for other banks: onboarding, mapping, validation, approval, events signed with HMAC-SHA256, per-bank keys and roles.
- **169 automated tests**, offline (no Fabric or Azure OpenAI call), run by GitHub Actions on every push. Public demo: hyson-detector.streamlit.app.
- **No secret in Git.** Keys live in a git-ignored file or the host's secrets store; Fabric access uses Entra ID, with no stored credential.

05 The detection model

5.1 Data

Item	Value
Transactions	**1,908,228** (Jan 2025 → Sep 2026)
Customers	3,500, in 18 cities
Digital-banking events	781,565 (logins, devices, beneficiaries, password and phone changes)
Legacy alerts with verdicts	32,065
Fraud rate	≈ 0.55 % (heavily imbalanced; real-world rates are lower)

Split by time, never at random. Train January 2025 → May 2026 (1,542,647 rows, 8,494 frauds); validation June–July 2026 (177,953 rows, 1,013 frauds); test August–September 2026 (187,628 rows, labels never used or shown). Fraud types in validation: stolen contactless card 406 · discreet online 264 · ATM skimming 182 · account takeover 61 · card testing 54 · cloned card abroad 46.

5.2 Features: 73, computed on the past only

Family	Examples
Amount vs habits	ratio and z-score to the customer's own history, above previous maximum, share of income, share of card limit
Velocity	operations, declines and micro-amounts on the card in 10 min, 1 h, 24 h
Novelty	first time with this merchant, category, device, beneficiary, IP country, city or country
Geography	distance and implied speed since the previous located operation (impossible travel)
Channel and security	magnetic stripe, contactless near the ceiling, online without 3-D Secure, domestic card abroad
Digital banking	failed logins, device or beneficiary added, password reset, phone change in the last 24 h
Graph	frauds on the same merchant or device in 30 days, counting **only frauds already reported** before the operation
Profile and calendar	age, income, tenure, KYC risk, payday, Ramadan, night

Never used: gender, identifiers, the fraud scenario, the report date (except "already reported before"), legacy analyst decisions. Three target-encoded risk columns that leaked future labels were found and removed; an assertion keeps them out.

5.3 Models compared

Model	PR-AUC	Top-1 % precision	Top-1 % recall
LightGBM (champion)	**0.646**	**40.4 %**	**70.9 %**
Rank blend	0.654	40.3 %	70.8 %
LambdaRank (one query per day)	0.647	39.9 %	70.0 %

Model	PR-AUC	Top-1 % precision	Top-1 % recall
HistGradientBoosting	0.640	39.9 %	70.1 %
19 explainable rules	0.014	2.8 %	4.9 %

The blend wins by less than a point, which is noise. LightGBM is kept because it is the model that produces the SHAP explanations: **the score and its explanation come from one model.** Settings: 600 trees, learning rate 0.03, 63 leaves, native categoricals; all frauds plus 10 % of legitimate operations, re-weighted to the true ratio.

5.4 From score to recommendation

The alert threshold is the **top 1 % of each period** (1,876 alerts on the test period), split into ELEVE (top 30 % of alerts) and MOYEN. A documented rule, computed in code, recommends:

Condition	Recommendation	Measured precision (validation)
Percentile ≥ 0.998 and ≥ 2 concordant signals	BLOQUER, high confidence	**94.1 %** (357 alerts)
ELEVE and ≥ 2 signals	BLOQUER, medium confidence	86.7 % (ELEVE band, 533 alerts)
ELEVE otherwise, or MOYEN	ESCALADER	20.5 % (MOYEN band, 1,246 alerts)
Below the alert threshold	VALIDER	—

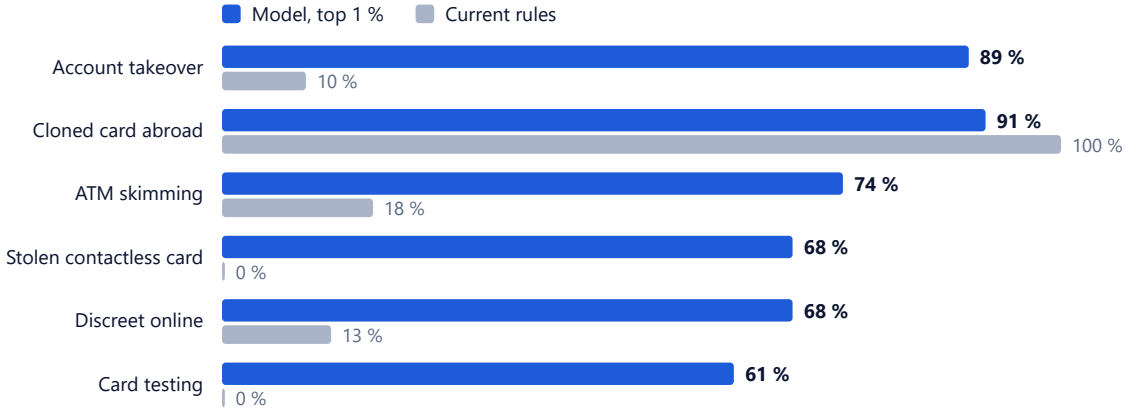
06 Results and robustness

6.1 Headline results

Validation June–July 2026	Alerts	Frauds caught	False alerts	Precision	Recall
Current rules engine	3,338	117	3,221	3.5 %	11.5 %
Model, same alert volume	3,338	787	2,551	23.6 %	77.7 %
Model, top 1 % (analyst capacity)	**1,779**	**718**	**1,061**	**40.4 %**	**70.9 %**
Model, top 0.5 %	890	590	≈ 300	66.4 %	58.2 %

In value, the model stops **67.7 %** of the money at risk against **22.1 %** for the rules. **PR-AUC 0.646** against 0.014. If a team can only review about 1,000 alerts in two months, the top 0.5 % keeps 58 % recall at 66 % precision.

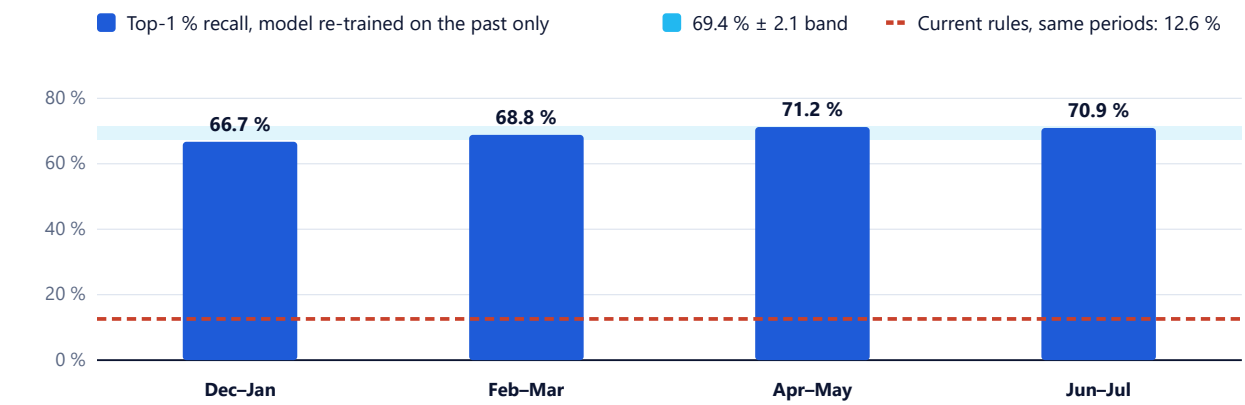
6.2 By fraud type



Recall by fraud type, validation June–July 2026. The legacy foreign-country rule is kept for cloned cards abroad.

The model beats the rules on five of the six fraud types. For cloned cards used abroad, the legacy foreign-country rule catches 100 % and is kept alongside the model: a hybrid, not a replacement.

6.3 Stability over time



Rolling backtest: four re-trainings on the past only, each tested on the next two months.

Same features, sampling and settings as production, re-trained four times on the past only, each tested on the next two months: **top-1 % recall 69.4 % ± 2.1 points**, precision 39.1 % ± 3.9, PR-AUC 0.622 ± 0.036, fraud value caught 72.5 %. The current rules reach 12.6 % recall and 3.7 % precision on the same periods.

95 % confidence intervals (bootstrap of whole days, 1,000 resamples): recall 70.9 % [68.3–73.0] · precision 40.4 % [34.9–45.0] · PR-AUC 0.646 [0.610–0.675] · false alerts avoided 2,160 [1,971–2,335].

6.4 Fairness

Group	Alert rate	Precision	Recall
Segments	0.88–1.12 %	30 % (retirees) → 46 % (premium)	61 % (retirees) → 76 % (professionals)
Age bands	0.95–1.09 %	35 % (60+) → 42 % (30–44)	65 % (18–29) → 75 % (30–44)

Legitimate customers wrongly alerted: the gap between the most and least disturbed group is **1.4×** across segments and **1.2×** across age bands for the model, against **2.8×** and **1.5×** for the current rules. Retirees and over-60s get lower precision and recall: shown openly, monitored, and a candidate for a dedicated threshold. Gender is neither used nor reported.

07 The AI layer

7.1 Fraud copilot

On each transaction page the analyst asks questions such as *"Why is this transaction suspicious?"*, *"Is it consistent with this customer's habits?"* or *"Are there related cases?"*. The copilot calls **10 read-only tools** over the Gold tables (alert queue, transaction, customer profile, comparison with habits, history, digital events, related cases across the bank, KPIs, person search, and decision recording on explicit request only) and answers in five parts: verdict, main factors, context, reassuring elements, recommendation.

- **Token discipline.** The app detects the type of question and fetches the facts itself, so most answers take **one** LLM round; the static prompt is cached; JSON is trimmed item by item. Measured: about **3,300 tokens and 7 seconds per new question** (down from 5,200 and 12), which is under one US cent at the gpt-5-mini list prices used in the app. Demo answers are pre-computed.
- **Guardrails.** Unknown IDs get "not found"; off-topic questions get a one-line refusal; prompt-injection text in the data is removed before the model sees it; an offline mode built on the same tools keeps the demo working without a key.
- **Evaluation.** 18 golden questions (the brief's questions word for word plus unknown ID, off-topic and injection): **18/18 pass**, 99.5 % of cited numbers found in the tool outputs.

7.2 AI assistant: ask the data

The first page is a single question box. Any question in French, English or Tunisian dialect becomes **one SQL query** on the star schema, which is parsed and checked before it runs: a single SELECT, only the nine star tables, no file, function or external access, a 1,000-row cap, and groups of at least 5 customers. The answer summarises only the returned rows (numbers copied, never computed by the AI), with an automatic chart, the table, the SQL and follow-up questions.

7.3 Nazar, the work agent

Nazar is a floating "evil eye" on every page. **Commands run in code, with zero tokens:** *my tasks* (briefing), *next* (opens the most urgent alert), *top 10 yesterday*, *alerts to block*, *open / trial / replay TX…*, person search, *check the log*, *kpi*. *Block this one* only **prepares** the decision with the system recommendation and the measured fraud rate; the analyst must click Confirm. Anything else becomes a short conversation on Azure OpenAI with Nazar's persona, which sees only aggregates and suggests the exact command to run. In **Voice** mode a 3D Nazar robot follows the cursor, talks and listens (browser speech, free). A second brain, plain GPT-5 mini, offers general chat with no access to bank data.

7.4 Investigation agents

- **The Trial.** An AI prosecutor and an AI defence argue the same Fabric facts on a numbered evidence board; the judge, an Azure AI Foundry agent, grades each argument and lists contradictions and missing information. Every citation is checked in code; the analyst accepts or overrides the verdict.
- **Deep search** on medium alerts whose customer already had an alert (≈ 53 % fraud against 16 %), with an evidence board and one AI call to connect the dots.
- **Reconstitution** replays the hours before the alert as a road map: account access, operations, the alert, and the day-0 detection milestone.
- **Counter-inquiry** neutralises the top SHAP factors one by one: the alert is *robust* if it survives, *sensitive* if it flips.

7.5 AI reports

An employee asks for a report in plain language, for example *"detailed quarterly report on ATM fraud compared with the previous quarter, for management"*. The platform shows what it understood (type, reader, period, comparison, scope, language, confidentiality, sections; all editable), then **analyses before writing**: fixed SQL per figure, verification (period, duplicates, sums, independent recount, label coverage), rule-based insights and charts with source. Confirmed fraud and suspected fraud are never mixed; every statement is tagged (verified fact, ML prediction, AI interpretation, hypothesis, recommendation, limitation). A named employee approves; PDF, DOCX, HTML and XLSX are archived with their SHA-256, in French, English or Arabic.

08 Personal context and continuous investigation

8.1 "Understand the person before suspecting the card"

A bank executive gave us this principle, with an example: a retired customer paying abroad. Hyson's **personal context** asks whether an operation is coherent with *this* customer's life, using only facts the bank already holds: previous trips to the place, travel preparation (airline, agency or hotel in the last 90 days), a stay in progress, time since the last payment elsewhere (impossible travel), chip or magnetic stripe, amount against his own habits and income, his phone logged in from that country, a travel notice he declared. It never infers religion, health, family or reasons.

Each signal is weighed by its fraud rate measured on 2025 and **tested on 2026** (4,138 payments abroad never used to fit): **ROC-AUC 0.858**. Operations labelled *coherent trip* (1,577 operations at 1.1 % fraud) include **1,559 of the 3,802 false alerts** the current engine raises abroad; *barely coherent* operations show 42 % fraud. An amount never reassures on its own, and the page gives a respectful script for contacting the customer.

8.2 The case lives on after the alert

A rejected alert is not a forgotten one. The **investigation case** has a time slider (at the alert, +6 h, +24 h, +72 h) that shows only what was known at each moment, and reactivation triggers: new uses of the card, declines, failed logins, two places at once, rising amounts. It retrieves the **8 most similar past cases** from a behavioural memory of **16,302 cases** (Azure OpenAI embeddings plus model features), with their status as known at the time.

Demo case TX01631347: ESCALADER at the alert, then BLOQUER at +24 h, when the same card appears in Tozeur and Tripoli four minutes apart. The employee accepts, rejects, asks for more, adds context, escalates or records the customer's confirmation, with a written rationale in an append-only file.

8.3 Fraud intelligence for managers

The steering page drills down from a channel to card types, then to transactions and same-card sequences. Facts are computed in SQL (concentration, magnetic-stripe fallback, authentication events, rules vs model coverage, trend, missing data), and the AI classifies one to three patterns (cybersecurity, administrative, technical, customer, coordinated) with evidence, confidence, the next investigation and expected impact, always separating fact from hypothesis.

09 Universal banking integration

Hyson is not tied to one database. A new bank connects its data whatever its schema, in seven steps, through the app or the REST API:

1. **Connect** a source (SQLite, CSV, JSON-lines, Parquet).
2. **Discover** tables, keys and links, drawn as a graph; only statistics leave the module.
3. **Understand**: AI-assisted roles and field mapping with a confidence score (names 60 %, values 40 %), code translation (ISO 8583, EMV, channel codes), sensitive columns excluded; uncertain items become flags to confirm.
4. **Validate**: readiness score, every KPI executed, reconciliation with the bank's control totals, capability matrix (a module turns on only if its data exists).
5. **Approve and activate** with a named approval; a bank-specific model trained on its own past is approved separately and never retrained automatically.
6. **Operate**: generated dashboard, alert queue with reasons, the bank's own memory, signed webhooks.
7. **Audit**: every step in the bank's own hash chain.

Validated end to end on three simulated banks with deliberately different schemas:

Bank	Source	Recognised	Reconciliation	Modules	Model on later months
Banque Atlas	SQLite, French names, dd/mm/yyyy	4/4 table roles	13,207 ops · 111 frauds · exact	13/15	PR-AUC 0.18 (base rate 1.3 %)
Nordic Pay	6 CSV, camelCase, epoch ms, EUR cents, ISO 8583	6/6 roles	15,408 ops · 127 frauds · exact	15/15	PR-AUC 0.155 vs 0.025 for the bank's own score
Sahel Bank	one JSON-lines file, no labels	1/1	8,853 ops · exact	5/15	behaviour score only, labelled as such

Security by design: one folder, key set, model, memory and audit trail per bank; API keys stored hashed with admin / analyst / ingest roles; identifiers pseudonymised per bank (HMAC); sensitive columns never mapped and erased at activation; a key never reaches another bank.

10 Governance, security and compliance

10.1 Human in control

- The model never blocks anything; the copilot and Nazar never record a decision on their own.
- Every decision stores who decided, the system recommendation, the model and prompt versions, the time from opening the alert to deciding, and a comment. Overrides of the recommendation are visible.
- Blocks above 5,000 TND, or overriding a BLOQUER recommendation, are flagged for four-eyes approval in the message prepared for the bank's systems.

10.2 Audit trail

Decisions, approvals, mapping versions, model approvals and archived reports are chained with **SHA-256**: each entry contains the hash of the previous one, and an integrity check on the Journal page shows at once whether anything was edited. Copilot calls are logged with question, tools, tokens and latency.

10.3 Privacy

- No raw rows are ever sent to an LLM; models receive compact facts and aggregates.
- Ask-the-data answers only on groups of at least 5 customers.
- Personal context never infers religion, health or family.
- Gender is not a feature; identifiers are pseudonymised per bank.

Before production: a declaration to the national data-protection authority (in Tunisia, the INPDP, under organic law 2004-63, to be confirmed with the bank's legal team), and validation on real data.

10.4 Model risk management

A one-page model card (purpose, data, features never used, threshold, explanations, performance with confidence intervals, fairness, limits) is ready for a validation team. Monitoring covers drift of the score distribution (PSI), alert volume per day and analyst decisions returned to OneLake. A new model version replaces v1 only if it beats it on the same backtest.

11 Market and benchmarks

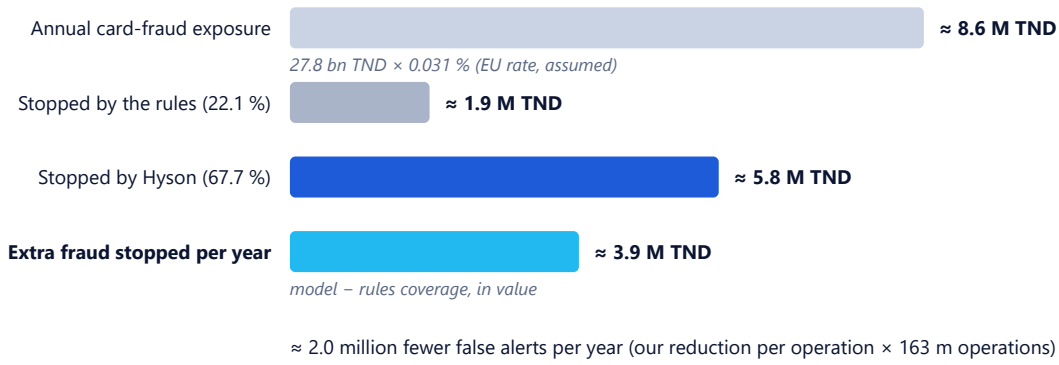
11.1 Our numbers next to public figures

The dataset is synthetic and deliberately richer in fraud than reality, so we compare **shapes** (shares and ratios), never absolute amounts. Every external figure has its source:

Public figure (source)	Value	Hyson data (measured)
Rule-based alerts that are false positives (PwC / McKinsey, industry estimate)	90–95 %	rules 96.5 % · model 59.6 %
Card fraud rate by value, EEA H1 2023 (EBA / ECB 2024)	0.031 %	0.28 %, deliberately higher
Cross-border share of card fraud value vs payment value (EBA / ECB 2024)	28 % vs 4 %	15.0 % vs 1.2 %, same pattern
Attempted unauthorised fraud stopped by UK banks (UK Finance 2025 / 2026)	67p → 70p per £1	fraud value stopped: model 67.7 % · rules 22.1 %
Payment fraud in the EU, 2024 (EBA / ECB, Dec 2025)	€4.2 bn (cards €1.33 bn, +29 %)	—
Card payments in Tunisia, 2024 (Banque Centrale de Tunisie)	163 m operations · 27.8 bn TND · 5.49 m cards · 3,305 ATMs	—

11.2 A Tunisian-scale scenario

Applying our validation rates to the real Tunisian card volume, with the EU fraud rate as an assumption (no public Tunisian rate exists):



Tunisian-scale scenario (BCT 2024 volumes, EU fraud rate as assumption). A scenario, not a measurement.

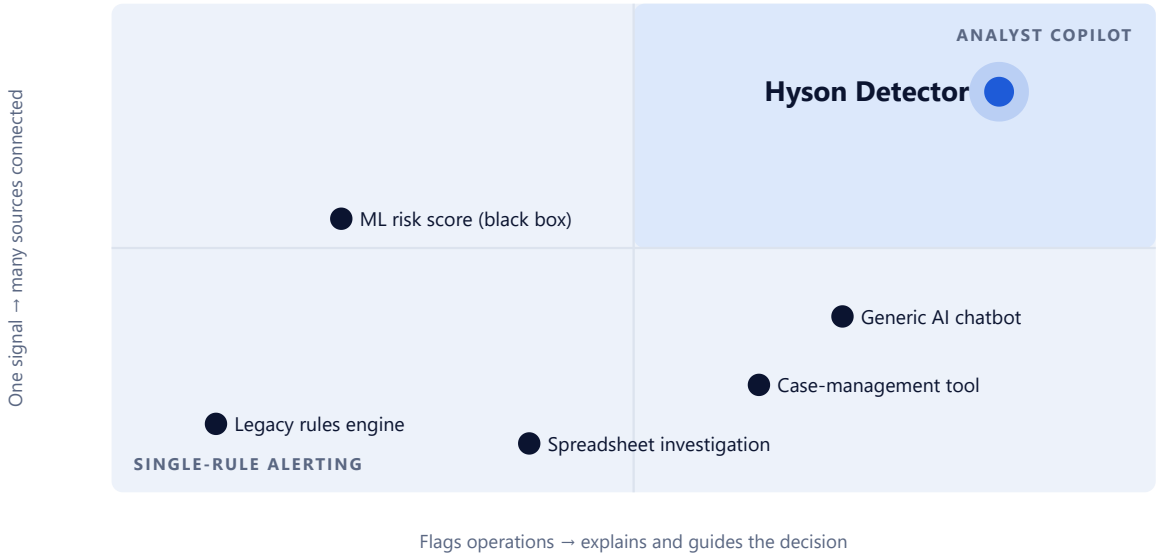
This is a **scenario, not a measurement**: the assumptions are editable in the app and meant to be discussed with a partner bank.

11.3 What the data says to a risk committee

- ATMs carry 10 % of the value but **25 % of the fraud**; online banking 7.5 % of value and **17.5 % of the fraud**.
- Magnetic stripe (0.81 %) and contactless (0.74 %) are about **3× riskier than chip** (0.26 %).

- Online payments **without 3-D Secure carry 1.8× more fraud** than with it, consistent with what EBA / ECB report about strong authentication.

11.4 Where Hyson fits



Positioning by category of tool, team assessment (not a vendor ranking).

Hyson does not replace the rules engine; it sits on top of it, tells the analyst which alert to open first and why, and keeps the trace auditors need.

12 Business model and go-to-market

12.1 Who it is for

- **Primary users:** fraud analysts and their team leads in Tunisian and North African retail banks and payment providers.
- **Secondary users:** risk managers and the risk committee (impact, Power BI, AI reports), compliance and internal audit (journal, model card), IT (integration API).
- **Value for the bank:** fewer false alerts per analyst, more fraud stopped in value, decisions that can be explained to customers and auditors, and customers spared unnecessary blocks.

12.2 Offer

Stage	What the bank gets
Pilot (3 months)	One fraud team, the bank's data mapped and approved, the model retrained on the bank's past, the copilot and the case file, weekly results against the current engine
Production	Real-time scoring, single sign-on, supervisor approval queue, AI reports, Power BI, monitoring and retraining under the bank's model-risk policy
Network	The same platform for several banks, each isolated, with shared typology statistics where the banks agree

Pricing will be set with the first partner from the pilot's measured results (false alerts avoided, fraud value stopped, analyst hours) rather than assumed in advance. The AI running cost is low and measured on every call (section 7.1).

12.3 Go-to-market

1. **Pilot with one bank**, sponsored by its fraud or risk director, on its historical data first, then on live alerts in parallel with the current engine.
2. **Publish the pilot's results** (same metrics as this report) to the bank's risk committee.
3. **Extend to payment providers and other banks** through the integration API, using the three-bank validation as proof that schemas differ but the method holds.

13 Roadmap

Horizon	Milestones
Now (hackathon)	Fabric pipeline, champion model, explained alert queue, copilot, assistant, Nazar, investigation agents, AI reports, universal integration, 169 tests, public demo
Pilot, months 1–3	Bank data mapped and approved; model retrained and validated on real data; weekly comparison with the current engine; random audit of about 0.1 % of non-alerted operations to measure missed fraud
Production, months 4–9	Real-time scoring with Fabric Eventstream and Eventhouse; Entra ID single sign-on; supervisor approval queue (four eyes); scheduled retraining pipeline; Azure AI Search for the bank's procedures; Azure AI Speech for the voice agent
Scale, months 10–18	Every Hyson page on each bank's own data; deep-learning challengers (sequence model, graph network, autoencoder) validated with the same 4-period backtest; multi-bank typology sharing

14 Risks and limits

We state them before anyone else does.

Limit	Mitigation
Synthetic, fraud-rich data (≈ 0.55 % fraud vs ≈ 0.1 % in reality): the model may have learned the generator	Retrain and validate on the pilot bank's real data before any decision
Fraud reported 11 days later on median: the latest weeks look falsely clean	In production, leave the last 30 days out of training (label maturity)
Feedback loop: only alerts get investigated, so missed frauds never become labels	Weekly random audit of about 0.1 % of non-alerted operations
Batch scoring, not at authorisation time	Real-time scoring is the first production milestone
Retirees and over-60s get lower precision and recall	Shown on the performance page; dedicated threshold and careful customer contact
Points of compromise are not predictive in validation (41 % vs 40 % precision)	Kept as an investigation lead only, not in the score
Behaviour score for banks without labels is weak alone (ROC-AUC 0.67–0.70)	It orders the review; it never decides
Analyst name typed, no SSO; four-eyes rule flagged but not enforced; core-banking message generated but not sent	Entra ID SSO and a supervisor queue in production
AI writing, agents' debate and mapping review tested offline only	Test with real calls during the pilot, with the same evidence checks
The 3D agent needs internet; voice uses the browser's speech services	Text mode works offline; Azure AI Speech in production

15 Team, tools and method

Team Hyson built Hyson Detector during the DSI Forum hackathon "AI Banking Copilot for Fraud Detection".

Area	Tools
Data and ML	Microsoft Fabric (lakehouse, notebooks, copy job, SQL endpoint, MLflow, data pipeline), PySpark, LightGBM, SHAP
AI	Azure OpenAI (gpt-5-mini, text-embedding-3-small), Azure AI Foundry agent, content safety filters
Analytics	Power BI (Direct Lake semantic model built as code)
Application	Python, Streamlit, FastAPI, DuckDB, sqlglot, Plotly, three.js
Engineering	GitHub, GitHub Actions (169 tests), GitHub Copilot; AI coding assistants declared in the repository

Method. Measure the current engine first, then improve on it with the same metrics; split by time and keep test labels out of reach; one model for the score and its explanation; numbers from code, words from the AI, decisions from people; state every limit.

Demo cases

Case	What it shows
TX01555470	Obvious account takeover → BLOQUER
TX01585580	Subtle card testing → BLOQUER
TX01575959	False positive avoided: a retired customer's 3,361 TND chip purchase in Tunis → VALIDER
TX01631347	Case that evolves: ESCALADER at the alert, BLOQUER at +24 h
TX01589761	Shared ATM (skimming)

Sources

EBA / ECB, *2024 Report on Payment Fraud* and December 2025 report · UK Finance, *Annual Fraud Report* 2025 and 2026 · Banque Centrale de Tunisie, card-payment statistics 2024 (via Managers.tn, November 2025) · PwC / McKinsey estimates cited by Fluxforce (2024) · DSI Forum hackathon brief and synthetic dataset. All Hyson figures are measured on the organisers' synthetic data in Microsoft Fabric.